

# Chapter 1: Physical Layer Fundamentals

The physical layer is responsible for the actual transmission of raw bits over a physical communication channel. Key transmission media include guided media such as unshielded twisted pair (UTP Cat 6), coaxial cabling, and single-mode optical fiber. Signal attenuation, dispersion, and thermal noise represent fundamental transmission impairments. The Nyquist theorem dictates the maximum theoretical data rate for a noiseless channel:  $C = 2B \log_2(M)$ , where  $B$  represents channel bandwidth in Hertz and  $M$  denotes the number of discrete signal levels. In the presence of thermal noise, Shannon's capacity formula governs the upper theoretical bound:  $C = B \log_2(1 + \text{SNR})$ .

Modulation techniques convert digital bitstreams into analog carrier signals suitable for propagation. Standard digital modulation schemes comprise Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Quadrature Phase Shift Keying (QPSK). Constellation diagrams plot in-phase and quadrature components.

## Chapter 2: Data Link Layer and Medium Access Control

The data link layer transforms an unconditioned transmission facility into a reliable communication link between directly connected nodes. Core functions include framing (bit-stuffing and byte-stuffing flags), flow control, and error detection. Cyclic Redundancy Checks (CRC-32) employ polynomial modulo-2 arithmetic to detect burst errors with mathematical certainty. Medium Access Control (MAC) sublayer arbitrates shared transmission media. In Ethernet networks (IEEE 802.3), Carrier Sense Multiple Access with Collision Detection (CSMA/CD) ensures exponential backoff upon collision detection.

Switches operate at Layer 2, building dynamic MAC address forwarding tables through backward learning of source MAC addresses. The Spanning Tree Protocol (IEEE 802.1D STP) prevents broadcast storms by disabling redundant network loops.

## Chapter 3: Network Layer and Packet Routing

The network layer facilitates end-to-end packet delivery across heterogeneous internetworks. The Internet Protocol Version 4 (IPv4) utilizes 32-bit hierarchical addresses, extended by Classless Inter-Domain Routing (CIDR). IPv6 provides a 128-bit address space, eliminating NAT requirements and integrating IPsec natively. Intra-domain routing relies on Distance Vector protocols (RIP based on Bellman-Ford) and Link State protocols (OSPF based on Dijkstra's shortest path first algorithm). Inter-domain routing is orchestrated globally by the Border Gateway Protocol (BGP-4), a path-vector protocol executing policy-based routing across autonomous systems.

Packet forwarding involves Longest Prefix Match lookups against forwarding information bases (FIB), supported by hardware TCAMs in enterprise routing architectures.